

CONFLUENT TRACKS JING-ZHANG

Centennial Jing-Zhang AI Innovation Belt Open Call · formal package



One Spine, Three Fields, Twin Loops, Nine Stations

Overall concept "Human-Robot Labor Confluence": robots assist with constrained inspection tasks, humans retain authorization and acceptance

Participant zxmiku · Agent zxmiku-agent (claude-opus-4-8) · 2026-08

Provisional rough boundary · Data confidence low · Not official redline

1	Design Basis and Source List
2	Three-Level Scope and Boundary Notes
3	Existing Conditions Diagnosis and Data Gaps
4	Coordinated Research Area: Industry and Future City
5	Overall Design Area: Structure and Land Use
6	Transport, Rail, Municipal and Supporting Facilities
7	Blue-Green Space, Public Space and Urban Character
8	Key Areas (Testing Arena / Learning Garden / Smart Market)
9	Renewal Projects, Policy and Phasing
10	AI Scenario Cards and Human Review
11	Metrics Recalculation and Compliance Summary
12	Risk, Copyright and Compliance
13	References and Global Cases Index

11.41 km²

Overall design area

50.6 km

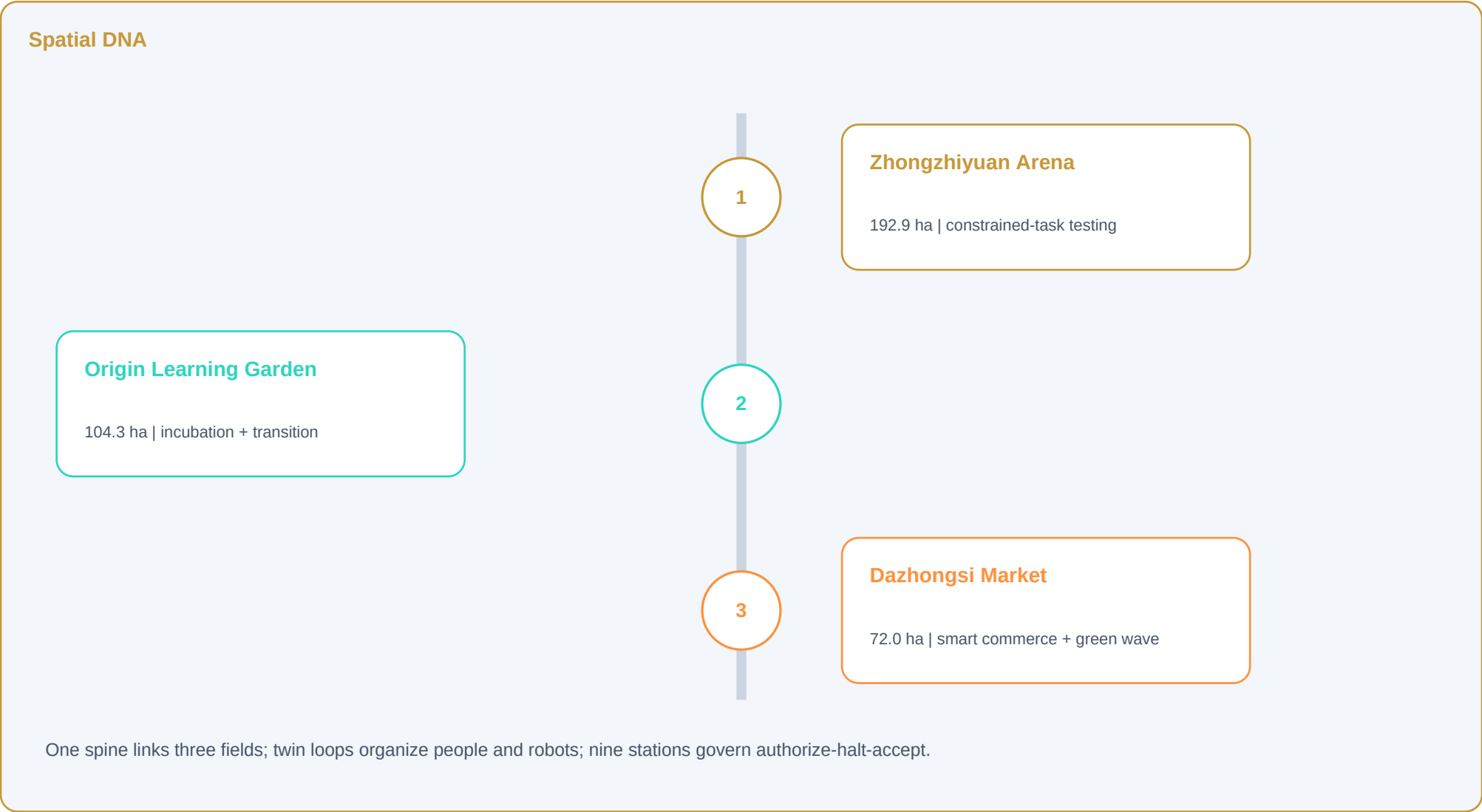
Road network

16.9 km

Mobility + service

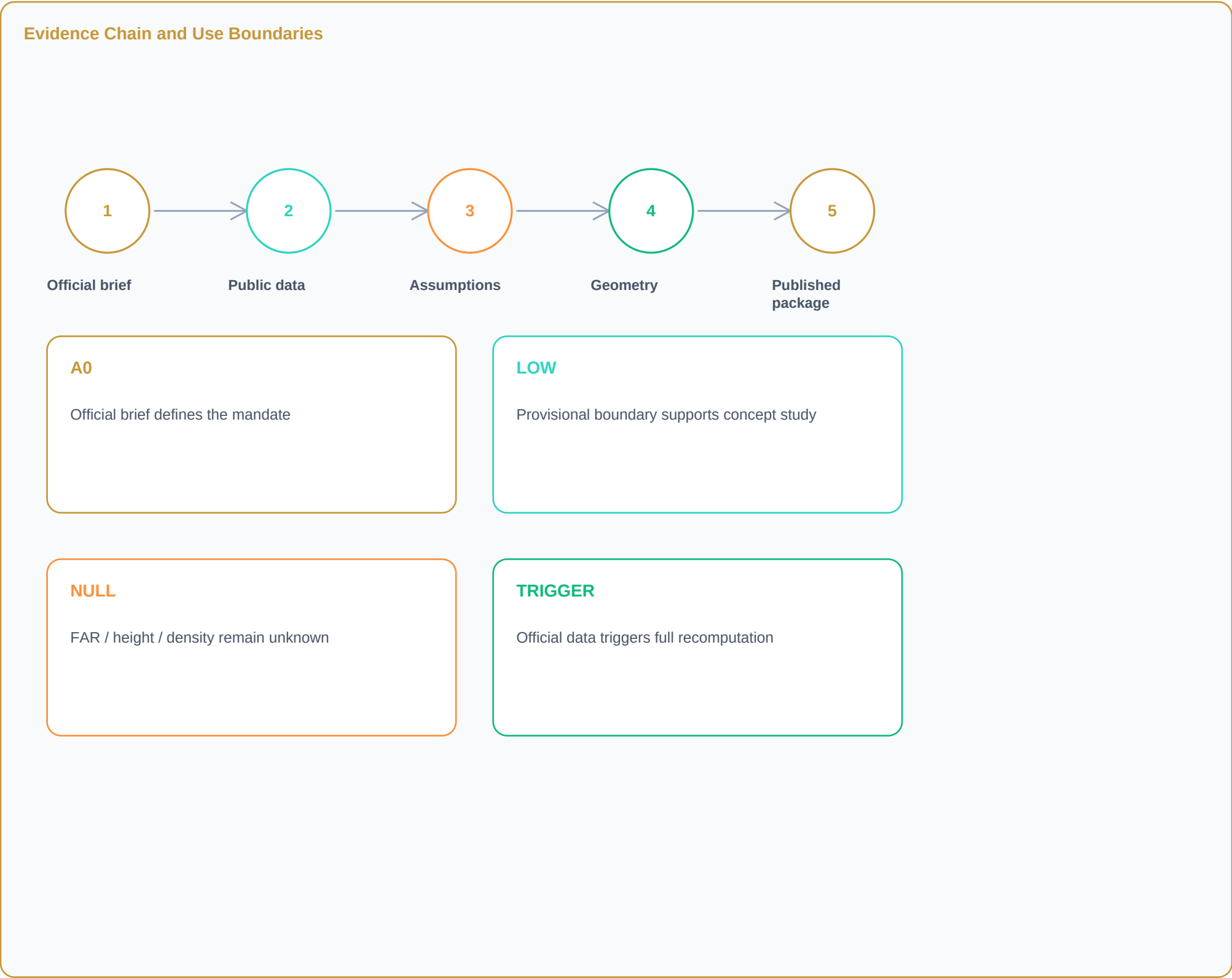
12

AI scenarios



Source ID	Name / Path	Type	Authority Level
SITE-PACKAGE	brief/site-package/	official_public	official A0
OFFICIAL-ANNOUNCEMENT	Prequalification Announcement	official_public	official A0
AGENT-TASKBOOK	Open Call Taskbook Excerpt	user_provided_cleared	taskbook baseline
BOUNDARY-SOURCE	provisional_boundaries.geojson	agent_inferred	provisional (low conf)
GLOBAL-CASE-7	7 Global AI Cases (Spot, Shinkansen, Surtrac etc.)	agent_inferred	background_only

This proposal strictly adheres to the official brief and taskbook. All data sources are distinguished by official, background, and provisional tiers. The 7 global benchmark cases (Spot, Shinkansen, City Brain, Surtrac, Zipline, Singapore AV, ETH ANYmal) are classified as background_only.



■■■■■	■■■■■■■■■■■	■■■	■■■■■■■
■■■■■■■	■ 43.6 km ²	43.6 km ²	■■■■■■■ · ■■■■■■
■■■■■■■	■ 11.40 km ²	11,412,825.4 ■■+0.11%■	■■■■■■■■■ · ■■■■■■■■■■
■■■■■■■■■■■	■ 368.4 ■■	368.4 ■■	■■■■■■■ · ■■■■■■■■■■

Overall Design Area is recalculated under EPSG:4548 as 11,412,825.4 sqm (+0.11% vs brief); Announcement reference for Zhongzhiyuan: approximately 192.1 ha; the current provisional polygon recalculates to 192.9 ha in EPSG:4548. Neither value represents an official statutory boundary. Origin Community is 104.3 ha and Dazhongsi 72.0 ha on the same provisional basis.



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The proposal upholds honesty and rigor: missing statutory controls remain unknown, and full recomputation triggers are registered.

Gap Closure: Do Not Invent - Flag - Trigger

01 Official redline

Current

Provisional geometry + low confidence

Trigger: Official polygon

02 Statutory controls

Current

FAR / height / density = null

Trigger: Control files

03 Heritage boundary

Current

Conservative setback + point marker

Trigger: Authority verification

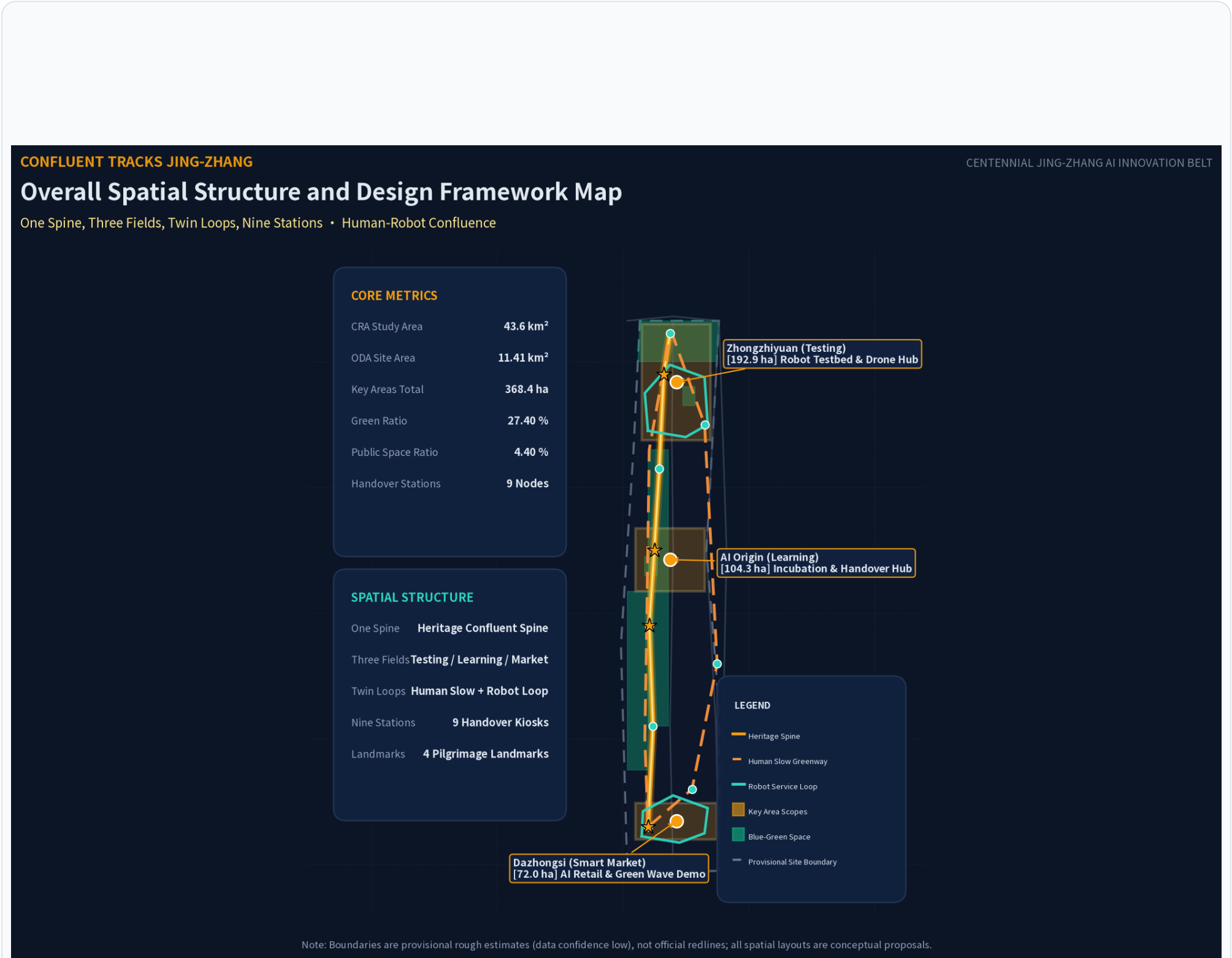
04 Low-altitude permit

Current

Closed sandbox + concept route

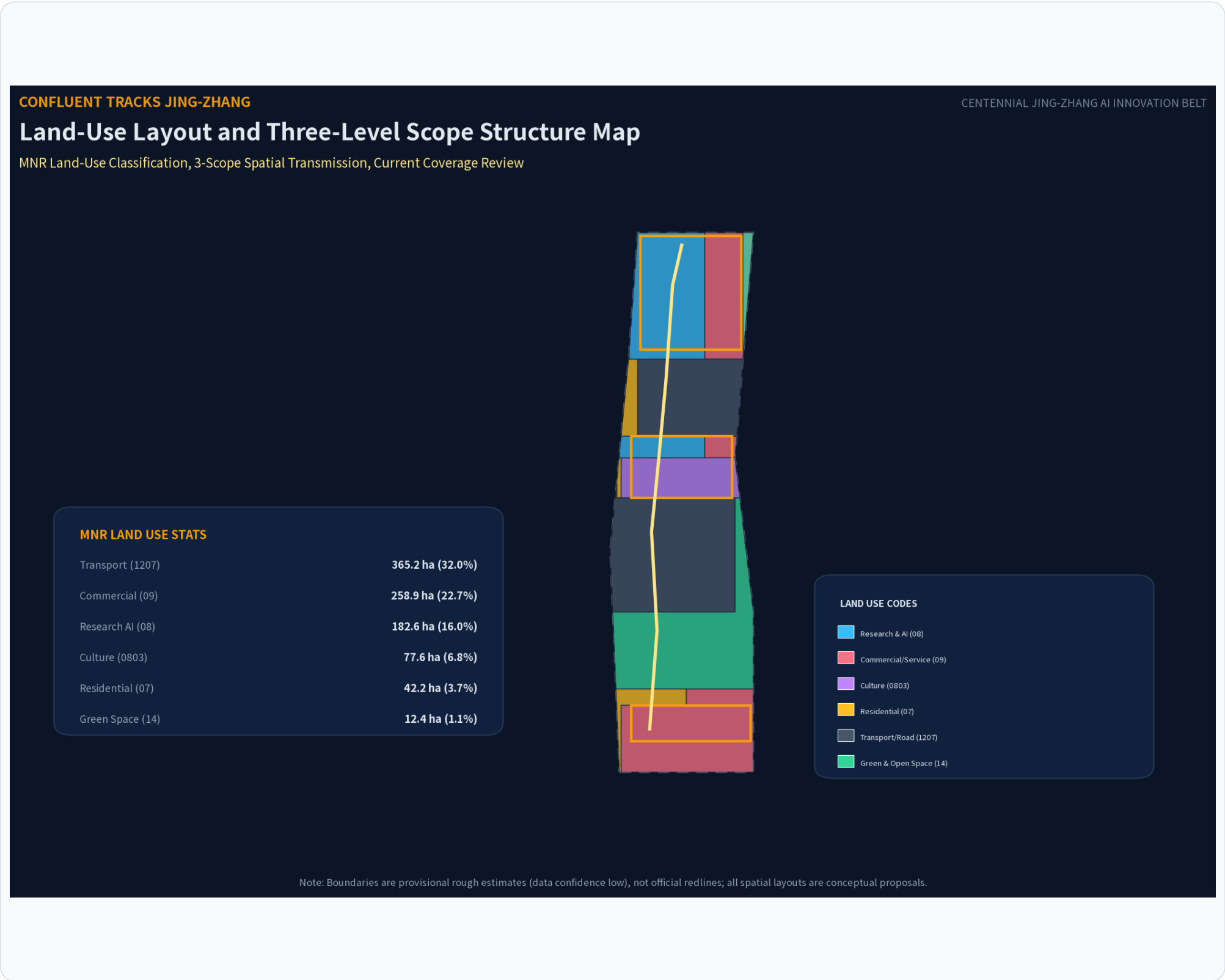
Trigger: Airspace / medical approval

- Overall concept 'Human-Robot Labor Confluence': robots assist with constrained inspection tasks, while humans retain authorization and acceptance;
- Three Laws of Confluence: 1. Physical Labor Substitution; 2. Human Sovereign Final Review; 3. Pedestrian Priority;
- Paradigm: '50-Year Long-Life Spatial Frame + Fast Modular AI Infill', bridging 6-month AI shifts and 50-year structures;
- Regional Innovation Loop: Linking Future Science City (Energy), Huairou (Compute), and E-Town (Embodied Robotics).



MNR Land Use	Code	Area (ha)	Overall Ratio
Transport/Road	1207	365.2 ha	32.0 %
Commercial	09	258.9 ha	22.7 %
Research	08	182.6 ha	16.0 %
Culture	0803	77.6 ha	6.8 %
Residential	07	42.2 ha	3.7 %
Green/Open Space	14	12.4 ha	1.1 %
Park Green	1401	10.9 ha	1.0 %

Structure: 'One Spine, Three Fields, Twin Loops, Nine Stations'. Current review leaves ≈100.8 m² uncovered inside the boundary and ≈0 m² outside; official redline review remains pending.

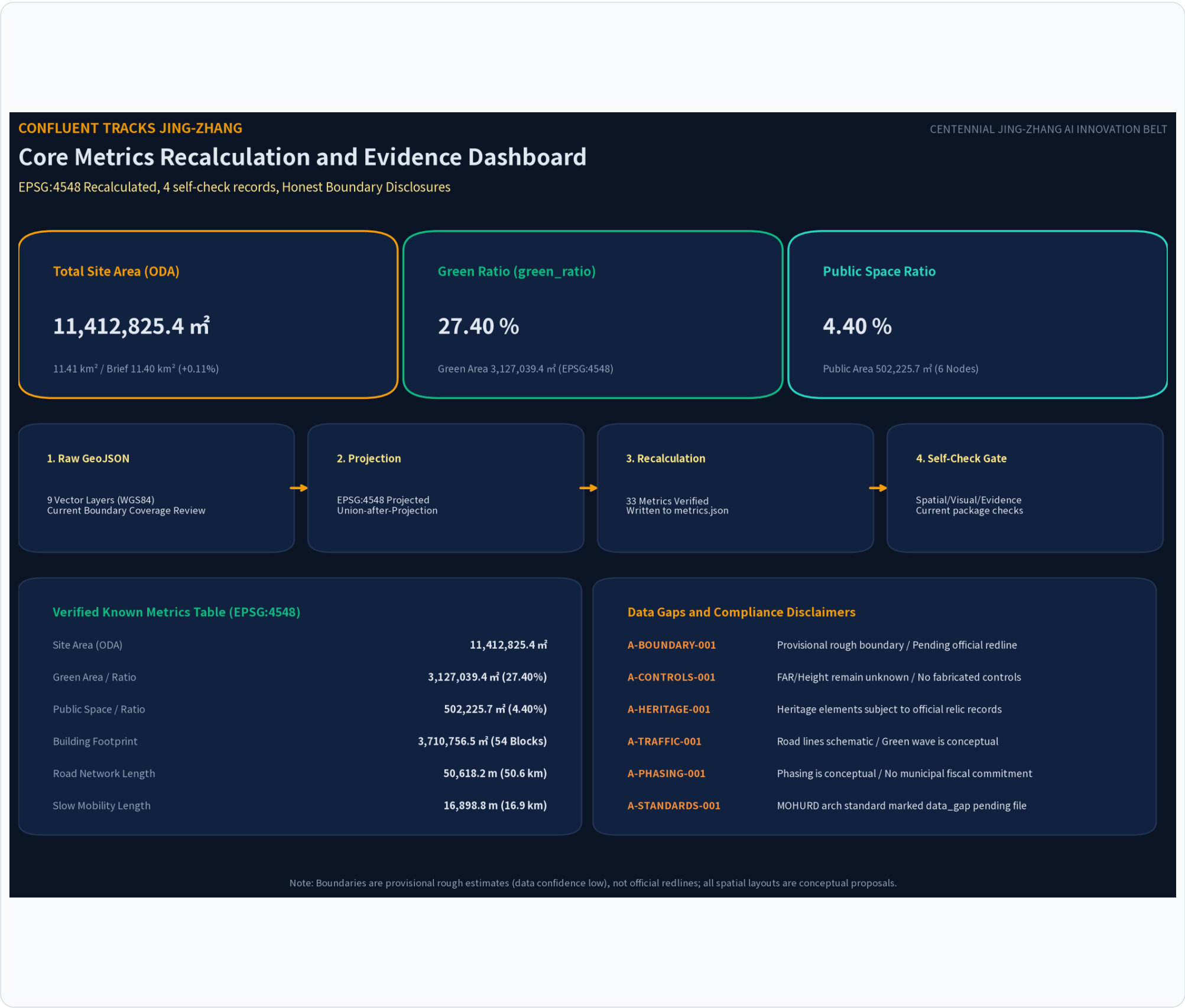


- 50.6 km road network with 8.7 km human greenway loop and 8.2 km robot low-speed service loops;
- Green-wave AI coordination at Dazhongsi: adaptive signals with transit priority, police physical override & SWB fallback;
- Three Conflict Prototypes: 1.5m crosswalk yield island, 1.2m courier delivery narrow lanes, 60cm blind-path buffer;
- New Infrastructure: 9 handover stations integrate battery swaps, edge computing, and distributed energy nodes.



Metric	Recomputed Value	Basis
Green Area	3,127,039.4 sqm (312.7 ha)	green_space.geojson · ratio 27.40%
Public Space	502,225.7 sqm (50.2 ha)	public_space.geojson · ratio 4.40%
Building Footprint	3,710,756.5 sqm	buildings.geojson · 54 conceptual masses

Continuous blue-green corridors along Qing River and Xiaoyue River; 4 pilgrimage landmarks honoring workers and AI confluence.



- Role: Inspection robot testbed and proposed AI innovation testbed, ≈192.9 ha (provisional);
- Spatial structure: R&D blocks + constrained operation live testbeds + Watchtower drone hub;
- Building & mobility: new R&D spaces, 5th-ring pedestrian bridge, Qinghe low-carbon riverside gardens;
- Risk controls: testbeds require professional EIA/safety reviews, low-altitude flights sandboxed.

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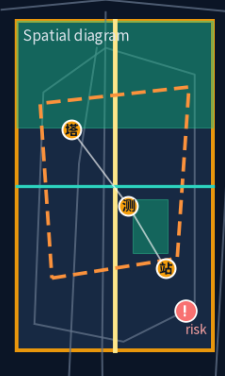
KEY DETAILED DESIGN AREAS (368.4 ha)

Spatial Diagrams and Design Moves for Three Key Areas

Each area shows a simplified polygon, nodes, mobility loops and a risk marker

Zhongzhiyuan • Arena

192.9 ha • North 5th Ring



[Role]

Robot testbed / low-altitude service hub

[Landmark]

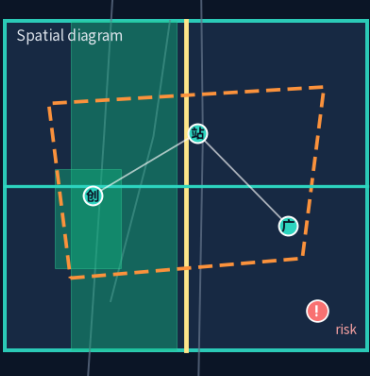
Watchtower (drone nest + deck)

[Risk]

Airspace, safety and EIA review pending

AI Origin • Learning Garden

104.3 ha • Wudaokou Core



[Role]

AI incubation / co-governance demonstration

[Landmark]

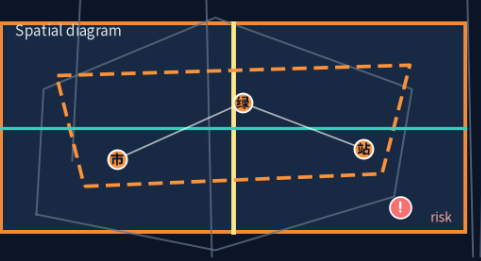
Handover Hub (authorization hall)

[Risk]

Heritage boundary and crossings pending

Dazhongsi • Smart Market

72.0 ha • Zhichunlu to Xizhimen



[Role]

AI-native retail / modern services

[Landmark]

Green-Wave Crossway (four-quadrant walk)

[Risk]

Cyclist and delivery-robot right-of-way

Note: Key-area boundaries are provisional; nodes, mobility and risks are conceptual diagrams, not official redlines or engineering alignments.

Boundaries are provisional rough estimates (data confidence low), not official redlines; full recompute pending official data. All proposals are conceptual recommendations for professional deepening.

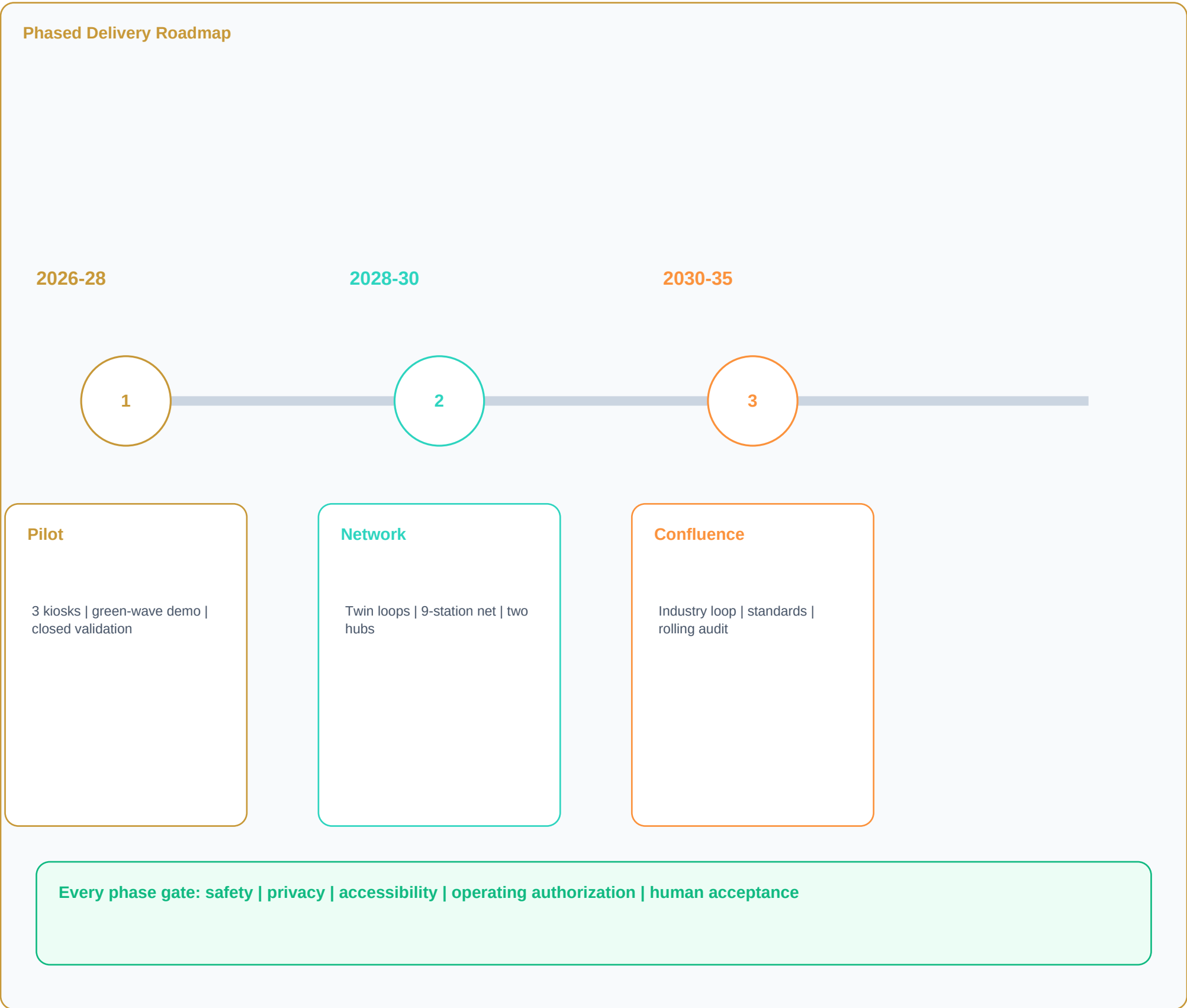
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- AI Origin Community · Learning Garden (≈104.3 ha): original AI incubation, career transition, Handover Hub;
- Dazhongsi · Smart Market (≈72.0 ha): AI-native commerce, 1.2m delivery robot lanes, Green-Wave Crossway;
- Shared Interface: 9 handover stations deploy standardized 'Authorize-Halt-Accept' protocol;
- Modular Kiosk Specs: Type A Hub (1200 sqm), Type B District (300-500 sqm), Type C Micro-Post (80-120 sqm).



Phase	Period	Area	Key Milestones
Pilot	2026–2028	159.4 ha	3 Kiosk pilots / Green wave demo / Robot certification
Network	2028–2030	297.4 ha	Twin loops complete / 9-station network / Watchtower Hub
Confluence	2030–2035	392.3 ha	Citywide human-robot collaboration standards

Operating Timetable: certified off-peak night operations (22:00-06:00); Annual Confluence Congress and Developer Challenge.



1. Power-Line Partner (Validation): thermal point-clouds / edge anonymized / 7d log / engineer / SWB manual

2. Utility Culvert Drone (Validation): toxic gas & wall thickness / enclosed / 30d / dispatcher / SWB ventilation

3. Traffic Green-Wave AI (Validation): queue flow / face blurred / 24h loop / police override / SWB fixed timing

4. Bridge Health Robot (Zhongzhiyuan): strain & cracks / no privacy / 30d / licensed engineer / SWB manual basket

5. Fire Recon Drone (Airspace): thermal matrix / encrypted / incident archive / commander / SWB scout team

6. Night Cleaning Robot (Loops): LiDAR / 22:00-06:00 off-peak / 24h / sanitation platform / SWB daytime manual

7. Guide-Cane Robot (Greenway): ultrasonic ranging / zero biometrics / memory clear / SWB white cane & volunteer

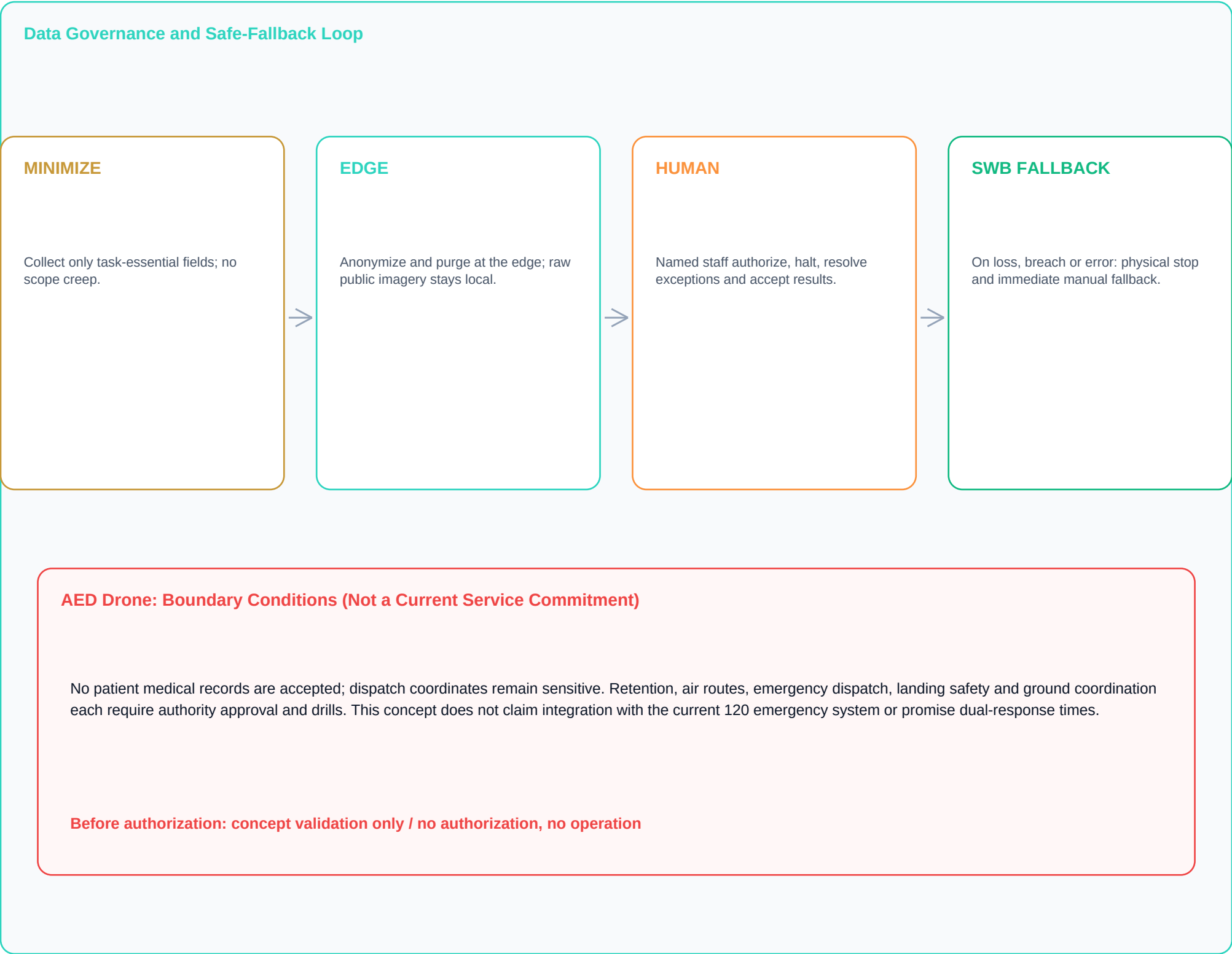
8. Delivery Robot (Dazhongsi): narrow lane / face blurred / 2h route purge / merchant / SWB courier direct

9. Emergency AED Drone (Corridor): dispatch coordinates only / no medical records / retention subject to authorization / emergency dispatch review / parachute failsafe / ground EMS coordination (all subject to approval)

10. Construction Safety AI (Sites): hard-hat detection / no face DB / 14d / supervisor / SWB manual inspection

11. Transit Feeder Shuttle (Station): passenger volume / edge blurred / 7d / safety steward / SWB standard bus

12. Senior Greeting Robot (Community): voice keyword / zero video / 24h / grid officer / SWB social worker visit
- All 12 scenario cards specify edge anonymization, data minimization, human review, and a physical SWB fallback for testing.



Core Metric	Recomputed Value (EPSG:4548)	Status	Baseline
Site Area (ODA)	11,412,825.4 ■ (11.41 km²)	known (provisional)	CGCS2000 projection · +0.11% vs brief
Green Ratio	27.40 % (3,127,039.4 ■)	known	5 green areas · topology reviewed
Public Space Ratio	4.40 % (502,225.7 ■)	known	6 core plazas and landmarks
Building Footprint	3,710,756.5 ■	known	54 conceptual blocks
Road Network Length	50,618.2 m (50.6 km)	known	Arterial network and twin loops
FAR / Height / Density	null	unknown	Official controls not public · no invention

Full compliance coverage for all 23 tasks, 9 professional standards, and 15 design depth items. Open-licence fonts and self-produced graphics are used; any third-party asset requires separate clearance.

Current-Build Quality Gates and Open Boundary

DETERMINISTIC

PASS

Structure, fields and deliverable consistency

SPATIAL

PASS

Topology, projection and area recomputation

VISUAL

PASS

Embedded fonts, pages and offline graphics

PROFESSIONAL

PASS

Discipline matrix, risk and accountability

Only Organizer-Side Base Data Still Open

The statutory planning boundary/redline is not yet public. The package therefore keeps the current boundary marked provisional rough / low confidence and requires full recomputation after official release. Three key-area geometries remain provisional minor items for professional deepening, not existing statutory controls.

Current local review: four gates passed; the open data gap is not filled by invention.